

Part 1: Single-Variable Differential Calculus

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Definition 1

Let $X \subset \mathbb{R}$. A **function** $f: X \rightarrow \mathbb{R}$ is a rule that assigns to each element $x \in X$ a unique value $y = f(x) \in \mathbb{R}$

$$f: X \rightarrow \mathbb{R}, x \mapsto y = f(x).$$

X is called the **domain** of f .

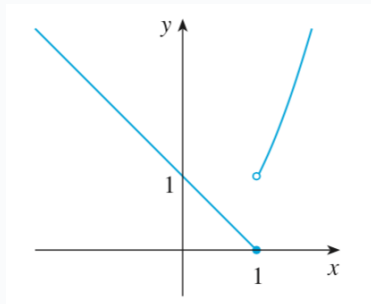
$f(X) = \{f(x), x \in X\}$ is called the **range** of f .

$\Gamma(f) = \{(x, f(x)) \mid x \in X\}$ is the **graph** of f .

x is an **independent variable**. y is a **dependent variable**.

Basic ways to represent a function

- Visually: by a graph.



- Numerically: by a data table. The table of values

x	1	2	3	4	5
$f(x)$	-1	0	6	1	-2

defines a function f . Determine its domain and range.

- Algebraic expression: by an explicit formula.

Piecewise-defined function, for example **Residential Electricity Price per Month**

Electricity Consumption (kWh)	Unit Price (VND/kWh)
Tier 1: 0 – 100 kWh	1.600
Tier 2: 101 – 300 kWh	2.000
Tier 3: Over 300 kWh	3.000

Let x be the electricity consumption (in kWh). Find a function $f(x)$ that calculates the payment amount.

$$f(x) = \begin{cases} 1,600 x, & \forall 0 \leq x < 100, \\ 160,000 + 2000(x - 100), & \forall 100 \leq x < 300, \\ 160,000 + 400,000 + 3000(x - 300), & \forall x \geq 300. \end{cases}$$

Definition 2

A function $f: X \rightarrow \mathbb{R}$ is said to be **bounded** on $D \subset X$ if there exists a constant M such that

$$\forall x \in D, -M \leq f(x) \leq M.$$

Example 1

- 1 $y = \sin x, x \in \mathbb{R}, y = \cos \frac{1}{x}, x \in \mathbb{R}^*$ are bounded by 1.
- 2 $y = \tan x, y = \cot x$ are unbounded on their domains.

Definition 3

Given a function $f: X \rightarrow \mathbb{R}$, an interval $I \subset X$.

i) f is called **increasing** on I if

$$f(x_1) \leq f(x_2) \text{ whenever } x_1 < x_2, \forall x_1, x_2 \in I.$$

ii) f is called **decreasing** on I if

$$f(x_1) \geq f(x_2) \text{ whenever } x_1 < x_2, \forall x_1, x_2 \in I.$$

iii) f is **strictly** increasing/decreasing if the strict inequality occurs.

① $y = \sin x, x \in [0, \frac{\pi}{2}]$ is an increasing function.

② $y = \sin x, x \in [\frac{\pi}{2}, \pi], y = \cot x, x \in (0, \pi)$, are decreasing functions.

Definition 4

A function $f: X \rightarrow \mathbb{R}$ is called an *even function* if

- X is symmetric about 0.
- $f(-x) = f(x)$ for all $x \in X$.

A function $f: X \rightarrow \mathbb{R}$ is called an *odd function* if

- X is symmetric about 0.
- $f(-x) = -f(x)$ for all $x \in X$.

Graphs of even functions are symmetric about Oy .

Graphs of odd functions are symmetric about the point O .

Example 2

- $\sin x, \tan x, \cot x$ are odd functions. $\cos x$ is an even function.
- $y = x^{2n}$ are even functions, $y = x^{2n+1}$ are odd functions, $n \in \mathbb{N}^*$.

Example 3

Determine whether the following functions are odd or even functions.

- 1 $f(x) = \sqrt[3]{(1-x)^2} + \sqrt[3]{(1+x)^2}$. Domain: $\mathbb{R}$.
 $f(-x) = \sqrt[3]{(1-(-x))^2} + \sqrt[3]{(1-x)^2} = f(x)$, so $f(x)$ is an even function.
- 2 $f(x) = \sin x + \cos x$. Domain: $\mathbb{R}$.
 $f(-x) = -\sin x + \cos x \neq f(x)$, $f(-x) \neq -f(x)$. $f(x)$ is neither even nor odd.

Definition 5

A function $f: X \rightarrow \mathbb{R}$ is said to be **periodic** if there exists a $p > 0$ such that

- for all $x \in X$, $x + p \in X$,
- $f(x) = f(x + p)$ for all $x \in X$.

The smallest possible value of p (if exists) is called the **period** T of f .

In literature, such p is also called "period", then T is called **fundamental/primitive/basic period**.

- 1 $y = \sin x$, $y = \cos x$ are periodic functions with period 2π .
- 2 $y = \tan x$, $y = \cot x$ are periodic functions with period π .
- 3 $y = \sin 2x + \cos 3x$ is a periodic function whose period is the least common multiple of π and $\frac{2\pi}{3}$, $T = 2\pi$.

Definition 6

Hyperbolic sine: $\sinh x = \frac{e^x - e^{-x}}{2}.$

Hyperbolic cosine: $\cosh x = \frac{e^x + e^{-x}}{2}.$

Hyperbolic tangent: $\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$

Hyperbolic cotangent: $\coth x = \frac{e^x + e^{-x}}{e^x - e^{-x}}, x \neq 0.$

New functions from old functions

- Translating graphs.
- Stretching or compressing transformations.
- Reflection through an axis.
- Taking the absolute value.
- Algebraic operations: sum, difference, product, quotient.

Definition 7

Let X, Y be subsets of $\mathbb{R}$ and $f: X \rightarrow Y$, $g: Y \rightarrow \mathbb{R}$ be two functions. Then the rule assigning x to $g[f(x)]$ is called the **composition** of g and f

$$g \circ f: X \rightarrow \mathbb{R},$$
$$x \mapsto z = g[f(x)].$$

Example 4

Let $f(x) = x^3 + 1$, $g(x) = (x + 1)^3$.
 $f \circ g(x) = (x + 1)^9 + 1$ **and** $g \circ f(x) = (x^3 + 2)^3$.
In general, $f \circ g \neq g \circ f$.

Definition 8

Let $f: X \rightarrow Y$ be a bijection. Then to each element $y \in Y$, there exists a unique element $x \in X$ such that $y = f(x)$. Therefore, $y \mapsto x$ determines a function

$$g: Y \rightarrow X,$$

$$y \mapsto x, y = f(x).$$

g is called the *inverse function* of f , g is also denoted by $f^{-1}(x)$.

Note: Domain of f = range of g ; range of f = domain of g .

$$f^{-1}(x) \neq \frac{1}{f(x)}.$$

Example 5

Find the inverse function of $y = \frac{2x-1}{x+3}$.

- Domain $x \neq -3$.

- For $x \neq -3$, we have

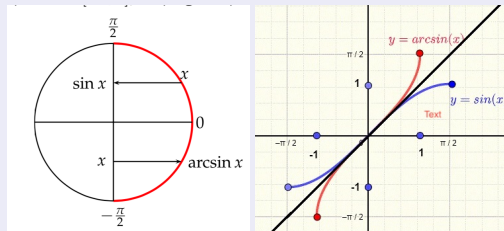
$$\begin{aligned} y = \frac{2x-1}{x+3} &\Leftrightarrow 2x - 1 = y(x + 3) \\ &\Leftrightarrow x(2 - y) = 3y + 1 \\ &\stackrel{y \neq 2}{\Leftrightarrow} x = \frac{3y+1}{2-y}. \end{aligned}$$

- Hence, the inverse function of $y = \frac{2x-1}{x+3} : \mathbb{R} \setminus \{-3\} \rightarrow \mathbb{R} \setminus \{2\}$ is $y = \frac{3x+1}{2-x}$.

Example 6 (Inverse trigonometric functions)

❶ The function $y = \sin x$ does not have an inverse.

However, $y = \sin x: \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$ is a bijection, this function has an inverse $\arcsin x: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

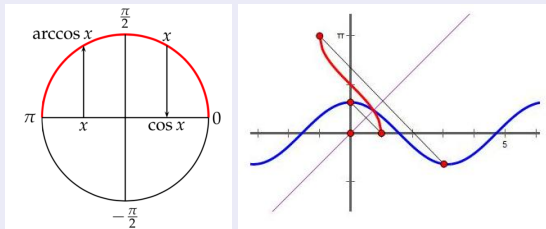


Hình: The graphs of $\sin x$ and $\arcsin x$.

$y = \arcsin x$ is an odd and increasing function.

Example 7 (Inverse trigonometric functions)

② $y = \cos x: [0, \pi] \rightarrow [-1, 1]$ is a bijection, its inverse is $\arccos x: [-1, 1] \rightarrow [0, \pi]$.



Hình: The graphs of $\cos x$ and $\arccos x$.

$y = \arccos x$ is a decreasing function.

Example 8 (Inverse trigonometric functions)

③ $y = \tan x: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ is a bijection, its inverse is the function $\arctan x: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

④ $y = \cot x: (0, \pi) \rightarrow \mathbb{R}$ is a bijection, its inverse is the function $\operatorname{arccot} x: \mathbb{R} \rightarrow (0, \pi)$.

$y = \arctan x$ is increasing, $y = \operatorname{arccot} x$ is decreasing.

Remark 1.1

The graphs of f and g are symmetric about the line $y = x$.

Indeed, $(x, f(x)) = (g(y), y) \in \Gamma(f); (y, g(y)) \in \Gamma(g), (g(y), y), (y, g(y))$ are symmetric about O .

Essential functions: exponential, logarithmic, power, trigonometric and inverse trigonometric functions.

$$a^x, \log_a x, x^\alpha,$$

$$\sin x, \cos x, \tan x, \cot x,$$

$$\arcsin x, \arccos x, \arctan x, \operatorname{arccot} x.$$

Elementary functions are *functions of one variable* built from essential functions using the elementary operations (sum, subtraction, product, division), composition of functions.

Notation:

$$\mathbb{R}_+ = \{x \in \mathbb{R} \mid x \geq 0\}, \mathbb{R}^* = \{x \in \mathbb{R} \mid x \neq 0\}.$$

- ❶ $f(x) = a^x$, $0 < a \neq 1$, the domain $\mathbb{R}$ and the range $\mathbb{R}_+^*$.
- ❷ $f(x) = \log_a x$, $0 < a \neq 1$, the domain is $\mathbb{R}_+^*$.
- ❸ $f(x) = x^\alpha$ has domains that are dependent on $\alpha \in \mathbb{R}$.
 - ▶ When $\alpha \in \mathbb{N}$, the domain is $\mathbb{R}$.
 - ▶ When $\alpha \in \mathbb{Z}_-$, the domain is $\mathbb{R}^*$.
 - ▶ When $\alpha = \frac{1}{p}$, $p \in \mathbb{N}^*$, the domain is $\mathbb{R}_+$ when p is even, and is $\mathbb{R}$ when p is odd.
 - ▶ When $\alpha \in \mathbb{R}$, in general the function is defined for $x > 0$.

A mathematical model is a mathematical description of a real world phenomenon by means of a function or an equation.

Mathematical modelling

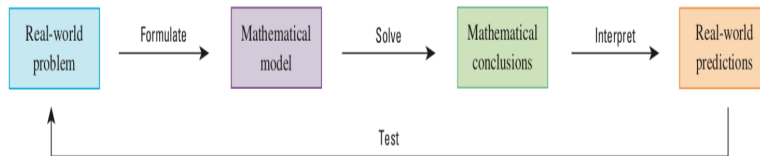


FIGURE 1 The modeling process

Definition 9

A number sequence is a function $f: \mathbb{N}^* \rightarrow \mathbb{R}$, $n \mapsto f(n) =: a_n$.

Notation $\{a_n\}_{n \geq 1}$.

Definition 10

The sequence $\{a_n\}$ is said to **converge to L as $n \rightarrow \infty$** iff for $\varepsilon > 0$, there exists $M(\varepsilon)$ such that $|a_n - L| < \varepsilon$ for all $n \geq M$.

We write $\lim_{n \rightarrow \infty} a_n = L$. $\{a_n\}$ is convergent.

- ① If the limit $\lim_{n \rightarrow \infty} a_n$ exists, then it is unique.
- ② If the sequence $\{a_n\}$ is convergent, then it is bounded, i.e. there exists $M > 0$ such that $|a_n| \leq M$, for all n .
- ③ If $a_n > A$ for all n , then $\lim_{n \rightarrow \infty} a_n \geq A$.
- ④ If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $a_n \neq 0$ for n large enough.

Proposition 2.1

Assume that $\lim_{n \rightarrow \infty} a_n = L_1$, $\lim_{n \rightarrow \infty} b_n = L_2$, then we have

- $\lim_{n \rightarrow \infty} (a_n \pm b_n) = L_1 \pm L_2.$
- $\lim_{n \rightarrow \infty} a_n \cdot b_n = L_1 \cdot L_2.$
- $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{L_1}{L_2}, (L_2 \neq 0 \Rightarrow b_n \neq 0 \text{ for } n \text{ large enough}).$

We also extend the concept of limit to the case $L = \infty$.

Theorem 2.1 (Cauchy's criterion)

The sequence $\{a_n\}$ is convergent iff it is a *Cauchy sequence*, i.e.

$$\forall \varepsilon > 0, \exists N_0(\varepsilon) \text{ such that } |a_m - a_n| < \varepsilon, \forall m, n \geq N_0.$$

Theorem 2.2 (Squeeze theorem)

Let $\{a_n\}, \{b_n\}, \{c_n\}$ be sequences that $a_n \leq b_n \leq c_n$ for $n \geq N_0$ and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$. Then $\lim_{n \rightarrow \infty} b_n = L$.

Theorem 2.3 (Monotone convergence theorem)

If $\{a_n\}$ is increasing and bounded from above, then it converges.

If $\{a_n\}$ is decreasing and bounded from below, then it converges.

Recall: $\{a_n\}$ is said to be *increasing (decreasing)* if $a_n \leq a_{n+1}$, $(a_n \geq a_{n+1})$, $\forall n \in \mathbb{N}$.

Test for convergence

❶ $a_n = \frac{\sin n}{\sqrt{n^2+1}}.$

❷ $a_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}.$

❸ $a_n = \left(1 + \frac{1}{n}\right)^n.$

1. We have: $-\frac{1}{\sqrt{n^2+1}} \leq \frac{\sin n}{\sqrt{n^2+1}} \leq \frac{1}{\sqrt{n^2+1}}, \forall n.$

Moreover, $\lim_{n \rightarrow \infty} \frac{-1}{\sqrt{n^2+1}} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2+1}} = 0.$

By squeeze theorem, $\lim_{n \rightarrow \infty} \frac{\sin n}{\sqrt{n^2+1}} = 0.$

2. $a_{2n} - a_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n} > \frac{1}{2n} + \frac{1}{2n} + \dots + \frac{1}{2n} = \frac{1}{2}.$

$\varepsilon = \frac{1}{4}, \forall N_0, n \geq N_0: |a_{2n} - a_n| > \varepsilon.$

By Cauchy's test, the sequence is not convergent.

3. $\{(1 + \frac{1}{n})^n\}$ is increasing and bounded from above by 3. It converges to $e = 2,7182\dots$

Two equivalent definitions of limit of a function

Definition 11

Let $f(x)$ be a function defined on (a, b) , possibly except at $x_0 \in (a, b)$. The *limit of $f(x)$ is a finite number L as x tends to x_0* , denoted by $\lim_{x \rightarrow x_0} f(x) = L$, iff

$$\forall \{x_n\} \subset (a, b) \setminus \{x_0\} : \lim_{n \rightarrow \infty} x_n = x_0 \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = L.$$

$$\Leftrightarrow \forall \varepsilon > 0, \exists \delta(\varepsilon, x_0) > 0 \mid 0 < |x - x_0| < \delta : |f(x) - L| < \varepsilon.$$

Remark 2.1 (Proof of nonexistence of $\lim_{x \rightarrow x_0} f(x)$)

There exist two sequences $\{x_n\}$ and $\{x'_n\}$ such that $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} x'_n = x_0$ BUT $\lim_{n \rightarrow \infty} f(x_n) \neq \lim_{n \rightarrow \infty} f(x'_n)$.

Example 9

- 1 $\lim_{x \rightarrow 1} (2x + 3) = 5$. Indeed, for $\varepsilon > 0$, choose $\delta = \frac{\varepsilon}{2}$: for all x that $|x - 1| < \delta$, we obtain

$$|(2x + 3) - 5| = 2|x - 1| < 2\delta = \varepsilon.$$

- 2 There does not exist $\lim_{x \rightarrow 0} \sin \frac{1}{x}$.

Choose $\left\{ \frac{1}{2n\pi} \right\}$ and $\left\{ \frac{1}{\frac{\pi}{2} + 2n\pi} \right\}$, which converge to 0.

But $\lim_{n \rightarrow \infty} \sin(2n\pi) = 0 \neq 1 = \lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{2} + 2n\pi\right)$.

Definition 12

Right hand limit at x_0 :

$$\lim_{x \rightarrow x_0^+} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \mid 0 < x - x_0 < \delta : |f(x) - L| < \varepsilon.$$

Left hand limit at x_0 :

$$\lim_{x \rightarrow x_0^-} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0 \mid 0 < x_0 - x < \delta : |f(x) - L| < \varepsilon.$$

We write: $f(x_0^+) = \lim_{x \rightarrow x_0^+} f(x)$, $f(x_0^-) = \lim_{x \rightarrow x_0^-} f(x)$.

Note: $\lim_{x \rightarrow x_0} f(x) = L \Leftrightarrow \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = L$.

Example 10

Find the limit $\lim_{x \rightarrow 0} f(x)$ (if exists), where

$$f(x) = \begin{cases} \frac{\sin 2x}{x}, & \text{when } x > 0, \\ 1 - \cos 3x, & \text{when } x \leq 0. \end{cases}$$

We have

$$f(0^+) = \lim_{x \rightarrow 0^+} \frac{\sin 2x}{x} = 2 \lim_{x \rightarrow 0^+} \frac{\sin 2x}{2x} = 2$$

$$f(0^-) = \lim_{x \rightarrow 0^-} (1 - \cos 3x) = 0.$$

$f(0^+) \neq f(0^-)$, hence, there does not exist $\lim_{x \rightarrow 0} f(x)$.

Definition 13

$$\lim_{x \rightarrow \infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists N(\varepsilon) > 0 \mid |x| > N(\varepsilon) : |f(x) - L| < \varepsilon.$$

- 1 The limit $\lim_{x \rightarrow x_0} f(x)$, if exists, is **unique**.
- 2 Squeeze theorem: assume that $f(x) \leq g(x) \leq h(x)$ for all $x \in (a, b) \setminus \{x_0\}$, and

$$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} h(x) = L.$$

Then $\lim_{x \rightarrow x_0} g(x) = L$.

Theorem 2.4

Assume that $\lim_{x \rightarrow x_0} f(x) = a$, $\lim_{x \rightarrow x_0} g(x) = b$. Then

- $\lim_{x \rightarrow x_0} [f(x) \pm g(x)] = a \pm b$.
- $\lim_{x \rightarrow x_0} f(x) \cdot g(x) = a \cdot b$.
- $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{a}{b}$ ($b \neq 0$, $g(x) \neq 0 \forall x_0 - \varepsilon < x < x_0 + \varepsilon$).

Note: x_0 stands for $x_0^\pm, \infty$.

Example 11

$$\lim_{x \rightarrow 0} x \cos \frac{1}{x} = 0.$$

The limit laws exclude the case $a = \infty$ or $b = \infty$ (infinite limits).

Definition 14

$$\lim_{x \rightarrow x_0} f(x) = \infty \Leftrightarrow \forall M > 0, \exists \delta > 0 \mid |x - x_0| < \delta : |f(x)| > M.$$

In cases of sum, difference, product, quotient involving infinite limits, we deal with indeterminate forms $\infty - \infty$, $0 \cdot \infty$, $\frac{0}{0}$, $\frac{\infty}{\infty}$.

For example, $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$; $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 3$.

Hence, we need to remove indeterminate forms.

Example 12 (Limit laws cannot be applied)

Find the following limits

$$a) \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 2}{x^2 - 1}$$

$$b) \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt{x+3} - 2}$$

$$c) \lim_{x \rightarrow 2} (2 - x) \tan \frac{\pi x}{4}$$

$$d) \lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x}}$$

$$a) \lim_{x \rightarrow 1} \frac{x^3 - 3x^2 + 2}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x^2 - 2x - 2)}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{x^2 - 2x - 2}{x + 1} = -\frac{3}{2}.$$

$$b) \lim_{x \rightarrow 1} \frac{\sqrt[3]{x} - 1}{\sqrt{x+3} - 2} = \lim_{x \rightarrow 1} \frac{\sqrt{x+3} + 2}{\sqrt[3]{x^2} + \sqrt[3]{x} + 1} = \frac{4}{3}.$$

$$c) \lim_{x \rightarrow 2} (2 - x) \tan \frac{\pi x}{4} = \lim_{x \rightarrow 2} \frac{2 - x}{\sin\left(\frac{\pi}{2} - \frac{\pi x}{4}\right)} \sin \frac{\pi x}{4} = \lim_{x \rightarrow 2} \frac{(2 - x) \frac{\pi}{4}}{\sin \frac{\pi}{4} (2 - x)} \frac{4}{\pi} = \frac{4}{\pi}.$$

Definition 15

A function $\alpha(x)$ is called an *infinitesimal* when $x \rightarrow x_0$ if $\lim_{x \rightarrow x_0} \alpha(x) = 0$.

Example 13

- x^m , $m > 0$, is an infinitesimal as $x \rightarrow 0$.
- $e^{\sqrt{x}} - 1$ is an infinitesimal as $x \rightarrow 0^+$.
- $\cot x$, $\cos x$ are infinitesimals as $x \rightarrow \frac{\pi}{2}$.

- ❶ If $\lim_{x \rightarrow x_0} f(x) = a$, we can write $f(x) = a + \alpha(x)$, where $\alpha(x)$ is an infinitesimal as $x \rightarrow x_0$.
- ❷ If $\alpha(x), \beta(x)$ are infinitesimals as $x \rightarrow x_0$ then $c_1\alpha(x) + c_2\beta(x)$, c_1, c_2 are constants, is also an infinitesimal as $x \rightarrow x_0$.
- ❸ If $\alpha(x)$ is an infinitesimal as $x \rightarrow x_0$, and $f(x)$ is **bounded** in $(x_0 - \varepsilon, x_0 + \varepsilon)$ then $\alpha(x)f(x)$ is also an infinitesimal as $x \rightarrow x_0$.

Example 14

$3x \sin \frac{1}{x} + 5 \ln(1 - 2x)$ is an infinitesimal as $x \rightarrow 0$.

Comparison of infinitesimals

Definition 16

Let $\alpha(x)$, $\beta(x)$ be infinitesimals as $x \rightarrow x_0$ and $\lim_{x \rightarrow x_0} \frac{\alpha(x)}{\beta(x)} = k$.

- If $k = 0$, $\alpha(x)$ is of *higher order than* $\beta(x)$, written as $\alpha(x) = o(\beta(x))$. Or $\beta(x)$ is of *lower order than* $\alpha(x)$.
If $k = \infty$, $\beta(x) = o(\alpha(x))$.
- If $k \neq 0, \infty$, $\alpha(x)$ and $\beta(x)$ are of the *same order*, written as $\alpha(x) = O(\beta(x))$.
In particular, $k = 1$, $\alpha(x)$ and $\beta(x)$ are said to be *equivalent*, written as $\alpha(x) \sim \beta(x)$.

For $m, n > 0$: x^m is of higher order than x^n as $x \rightarrow 0$ iff $m > n$.

Example 15

Compare the following two infinitesimals $\alpha(x) = \sin 2x$ and $\beta(x) = \ln(1 + x^2)$ as $x \rightarrow 0$.

Lemma 2.1 (Fundamental equivalent pairs of infinitesimals)

As $x \rightarrow 0$:

- $\sin x \sim x, \arcsin x \sim x, \tan x \sim x, \arctan x \sim x$.
- $\ln(1+x) \sim x, a^x - 1 \sim x \ln a$. *In particular, $e^x - 1 \sim x$.*
- $1 - \cos x \sim \frac{x^2}{2}$.
- $(1+x)^\alpha - 1 \sim \alpha x, \alpha \in \mathbb{R}$.

Example 16

Compare the following two infinitesimals $\alpha(x) = \sqrt[3]{1+7x^2} - 1$ and $\beta(x) = \arcsin 2x$ as $x \rightarrow 0$.

We can simplify an infinitesimal by the following rules (all the quantities below are infinitesimals in the process $x \rightarrow x_0$).

Proposition 2.2

- $\alpha(x) + o(\alpha(x)) \sim \alpha(x)$.
- If $\alpha(x) \sim \beta(x)$ *then* $\alpha(x)f(x) \sim \beta(x)f(x)$, ($f(x) \neq 0$).

In particular, if $\alpha(x) \sim \tilde{\alpha}(x)$, $\beta(x) \sim \tilde{\beta}(x)$, then

$$\lim_{x \rightarrow x_0} \frac{\alpha(x)}{\beta(x)} = \lim_{x \rightarrow x_0} \frac{\tilde{\alpha}(x)}{\tilde{\beta}(x)}.$$

Example 17

Find the following limits

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{(2^x - 1) \ln(1+2x)}{\sin^2 x}$$

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{\sqrt{1+3x} - \sqrt[3]{1+6x}}{x}$$

$$\textcircled{3} \lim_{x \rightarrow 0} \frac{x + \sin^3 x - 2 \tan^4 x}{e^{2x} - \cos x}$$

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{(2^x - 1) \ln(1+2x)}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{x \ln 2 \cdot 2x}{x^2} = 2 \ln 2.$$

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{\sqrt{1+3x} - \sqrt[3]{1+6x}}{x} = \lim_{x \rightarrow 0} \frac{(1+3x)^{\frac{1}{2}} - 1 - ((1+6x)^{\frac{1}{3}} - 1)}{x} = -\frac{1}{2}.$$

$$\textcircled{3} \lim_{x \rightarrow 0} \frac{x + \sin^3 x - 2 \tan^4 x}{e^{2x} - \cos x} = \lim_{x \rightarrow 0} \frac{x}{e^{2x} - 1 + 1 - \cos x} = \lim_{x \rightarrow 0} \frac{x}{2x} = \frac{1}{2}.$$

Remark 2.2

Be careful when simplifying a sum of the form $\alpha(x) + O(\alpha(x))$.

We know that $\tan x \sim \sin x \sim x$ as $x \rightarrow 0$.

BUT $\tan x - \sin x \sim \frac{x^3}{2}$ and $x - \sin x \sim \frac{x^3}{6}$ are not equivalent as $x \rightarrow 0$.

Definition 17

A function $A(x)$ is called an *infinity* as $x \rightarrow x_0$ if $\lim_{x \rightarrow x_0} A(x) = \infty$.

Example 18

- x^n , $n > 0$, are infinities as $x \rightarrow \infty$.
- x^n , $n < 0$, are infinities as $x \rightarrow 0$.
- $\tan x$ is an infinity as $x \rightarrow \frac{\pi}{2}$.

Note: $A(x)$ is an infinity as $x \rightarrow x_0$ iff $\frac{1}{A(x)}$ is an infinitesimal as $x \rightarrow x_0$.

Definition 18

Let $A(x)$, $B(x)$ be two infinities as $x \rightarrow x_0$ and $\lim_{x \rightarrow x_0} \frac{A(x)}{B(x)} = k$.

- If $k = 0$, $A(x)$ is said to be of **lower order than** $B(x)$ as $x \rightarrow x_0$, ($B(x)$ is an infinity of higher order than $A(x)$).

If $k = \infty$, $B(x)$ is said to be of lower order than $A(x)$.

- If $k \neq 0, \infty$, we say $A(x)$ and $B(x)$ are of the **same order**.

In particular, $k = 1$, we say $A(x)$ and $B(x)$ are **equivalent**, written as $A(x) \sim B(x)$.

If $m > n > 0$: x^m is of higher order than x^n as $x \rightarrow \infty$.

Functions $\ln x$, x^α , a^x are of increasing order, as $x \rightarrow \infty$, here $\alpha > 0, a > 1$.

Applications to eliminating indeterminate forms

We can simplify an infinity by the following rules (all the quantities below are infinities in the process $x \rightarrow x_0$).

Proposition 2.3

- ❶ If $\overline{A}(x)$ is an infinity of lower order than $A(x)$, then $A(x) + \overline{A}(x) \sim A(x)$.
- ❷ If $A(x) \sim \overline{A}(x)$, then $A(x)f(x) \sim \overline{A}(x)f(x)$, ($f(x) \neq 0$).

In particular, if $A(x) \sim \overline{A}(x)$, $B(x) \sim \overline{B}(x)$, then

$$\lim_{x \rightarrow x_0} \frac{A(x)}{B(x)} = \lim_{x \rightarrow x_0} \frac{\overline{A}(x)}{\overline{B}(x)}.$$

Example 19

$$\lim_{x \rightarrow \infty} \frac{x+2x^2+4x^5}{20x^2+x^{2022}} = \lim_{x \rightarrow \infty} \frac{4x^5}{x^{2022}} = 0.$$

Remark 2.3

Be careful when simplifying a sum of the form $A(x) + \overline{A}(x)$ and $A(x), \overline{A}(x)$ are of the same order.

For example, as $x \rightarrow \infty$: $\sqrt{x^2 + 4} - x$ is an infinitesimal; $\sqrt{x^2 + 4x + 1} - x$ is neither an infinity nor an infinitesimal.

Definition 19

Let $f(x)$ be defined on (a, b) , $x_0 \in (a, b)$.

- $f(x)$ is said to be **continuous at x_0** if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. x_0 is a point of continuity.
- If $f(x)$ is not continuous at x_0 , it is said to be **discontinuous at x_0** . x_0 is a point of discontinuity.
- $f(x)$ is **continuous on (a, b)** iff $f(x)$ is continuous at $x \in (a, b)$.

Definition 20

- Let $f(x)$ be defined on $(x_0 - \varepsilon, x_0]$, $f(x)$ is said to be *continuous from the left* at x_0 if

$$f(x_0^-) = \lim_{x \rightarrow x_0^-} f(x) = f(x_0).$$

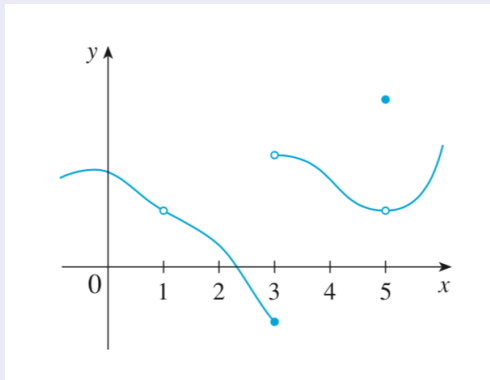
- Let $f(x)$ be defined on $[x_0, x_0 + \varepsilon)$, $f(x)$ is said to be *continuous from the right* at x_0 if

$$f(x_0^+) = \lim_{x \rightarrow x_0^+} f(x) = f(x_0).$$

Note: f is continuous at $x_0 \Leftrightarrow \lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) = f(x_0)$.

Example 20

- ① Elementary functions are continuous on their domains.
- ② Discontinuity from graph



Example 21

- 3 Determine whether the following function is continuous or not at $x = 0$

$$g(x) = \begin{cases} 2 - e^x & \text{if } x \geq 0, \\ \cos \pi x & \text{if } x < 0. \end{cases}$$

- 4 Determine a such that the function $f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0, \\ a & \text{if } x = 0, \end{cases}$ is continuous at 0.

- 5 Locate the discontinuity of the function and investigate the one sided continuity of $h(x)$ at these points,

$$\text{where } h(x) = \begin{cases} \frac{1 - \cos x}{\ln(1 + x^2)} & \text{if } x < 0, \\ \frac{1}{2} & \text{if } x = 0, \\ \tan x & \text{if } 0 < x < \frac{\pi}{2}, \\ x^2 + 1 & \text{if } x \geq \frac{\pi}{2}. \end{cases}$$

Definition 21

$f(x)$ is *continuous on $[a, b]$* iff $f(x)$ is continuous on (a, b) , **continuous from the right at a** and **continuous from the left at b** .

Definition 22

Let x_0 be a point of discontinuity of $f(x)$. x_0 is called

- a **jump discontinuity** if there exist finite limits $\lim_{x \rightarrow x_0^+} f(x)$ and $\lim_{x \rightarrow x_0^-} f(x)$.
- a **removable discontinuity** if $\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) < \infty$.
- an **infinite (essential) discontinuity** otherwise, i.e. at least one of $\lim_{x \rightarrow x_0^+} f(x)$ and $\lim_{x \rightarrow x_0^-} f(x)$ does not exist or is infinite.

Example 22

- $x = 0$ is a removable discontinuity of $f(x) = \frac{\sin x}{x}$.
- $x = 0$ is an essential discontinuity of $f(x) = e^{\frac{1}{x}}$.

Example 23

Classify the discontinuities of the functions

$$3 \quad y = \begin{cases} 3x^2 & \text{if } x < 0, \\ \frac{\ln(1+2x)}{\arctan x} & \text{if } x \geq 0. \end{cases}$$

$$4 \quad y = \frac{2}{1 - 2^{\frac{1}{x-1}}}.$$

Theorem 3.1

Let $f(x)$ and $g(x)$ be continuous at $x_0 \in (a, b)$.

Then $f(x) \pm g(x)$, $f(x) \cdot g(x)$ are also continuous at x_0 .

If $g(x_0) \neq 0$ then $\frac{f(x)}{g(x)}$ is also continuous at x_0 .

Theorem 3.2 (Continuity of composite functions)

Given $f(x): (a, b) \rightarrow (c, d)$, $g(x): (c, d) \rightarrow \mathbb{R}$. If $f(x)$ is continuous at $x_0 \in (a, b)$, $g(x)$ is continuous at $y_0 = f(x_0)$ then $g \circ f(x)$ is continuous at x_0 .

Application:

$$\begin{aligned} \lim_{x \rightarrow x_0} [f(x)]^{g(x)} &= \lim_{x \rightarrow x_0} e^{\ln(f(x)^{g(x)})} = \lim_{x \rightarrow x_0} e^{g(x) \ln f(x)} \\ &= e^{\lim_{x \rightarrow x_0} g(x) \ln f(x)}. \end{aligned}$$

Theorem 3.3

If $f(x)$ is continuous on $[a, b]$, then the graph of $f(x)$ has no break in it.

Theorem 3.4 (Weierstrass)

*If $f(x)$ is continuous on $[a, b]$ then it is **bounded, and attains its minimum and maximum** on $[a, b]$.*

Theorem 3.5

Let $f(x)$ be continuous on $[a, b]$. Then $f(x)$ attains all intermediate values between $f(a)$ and $f(b)$.

Corollary 3.1 (Intermediate Value Theorem)

Assume that $f(x)$ is **continuous on the closed interval** $[a, b]$ and $f(a) \cdot f(b) < 0$. Then the equation $f(x) = 0$ has at least one solution on (a, b) .

Example 24

Prove that the equation $x^6 - 4x + 2 = 0$ has at least two roots in the interval $(0, 2)$.

Example 25

Let $f(x): [0, 1] \rightarrow [0, 1]$ be a continuous function. Prove that the equation $f(x) = x$ has at least one solution in $[0, 1]$.

Theorem 3.6

If $f(x)$ is continuous on $[a, b]$, then $f(x)$ is uniformly continuous on $[a, b]$.

Definition 23

$f(x)$ is uniformly continuous on X iff

$$\forall \varepsilon > 0, \exists \delta(\varepsilon) > 0 : \forall x_1, x_2 \in X, |x_1 - x_2| < \delta : |f(x_1) - f(x_2)| < \varepsilon.$$

Recall: $f(x)$ is continuous at $x_0 \in X \Leftrightarrow \lim_{x \rightarrow x_0} f(x) = f(x_0)$,

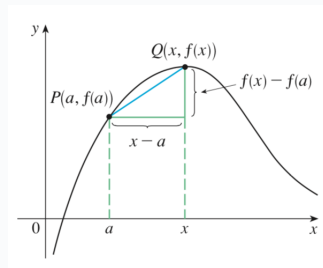
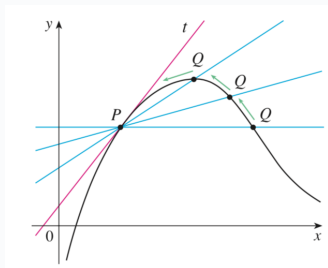
$$\forall \varepsilon > 0, \exists \delta(\varepsilon, x_0) > 0 : \forall 0 < |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon.$$

Remark 3.1

The uniform continuity is stronger than the continuity.

For example, $f(x) = \frac{1}{x}$ is continuous on $(0, 1)$, but it is not uniformly continuous on $(0, 1)$.

Geometrical illustration



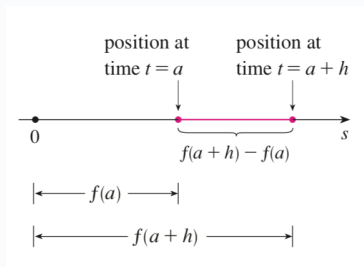
The slope of the secant PQ is

$$k_{PQ} = \frac{f(x) - f(a)}{x - a}$$

The slope of the tangent line to the graph of $f(x)$ at P is

$$k = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Physical illustration



The average velocity over the time interval $[a, a + h]$ is

$$v_m = \frac{f(a + h) - f(a)}{h}$$

The instantaneous velocity at $t = a$ is

$$v = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Definition 24

Let $f(x)$ be defined on (a, b) , $x_0 \in (a, b)$. $f(x)$ is said to *have a derivative at x_0* if there exists a finite limit

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} =: f'(x_0).$$

Another notation: set $x - x_0 = \Delta x$, we can write

$$\frac{df}{dx}(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta f}{\Delta x}$$

Δx : the increment in x , Δf : the corresponding increment in f .

Definition 25 (One-sided derivative)

- Let $f(x)$ be defined on $[x_0, x_0 + \varepsilon)$, $\varepsilon > 0$. *The righthand derivative at x_0 is*

$$f'_+(x_0) = \lim_{x \rightarrow x_0^+} \frac{f(x) - f(x_0)}{x - x_0}.$$

- Let $f(x)$ be defined on $(x_0 - \varepsilon, x_0]$, $\varepsilon > 0$. *The lefthand derivative at x_0 is*

$$f'_-(x_0) = \lim_{x \rightarrow x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

Note that $\exists f'(x_0) \Leftrightarrow \exists f'_+(x_0) = f'_-(x_0)$.

Definition 26

$f(x)$ is said to be *differentiable at $x = x_0$* if there exists $f'(x_0)$.

$f(x)$ *has derivative on the open interval (a, b)* if $f(x)$ has a derivative **at all points** $x \in (a, b)$.

$f(x)$ *has derivative on the closed interval $[a, b]$* if $f(x)$ has derivative on the open interval (a, b) , lefthand derivative at b , and righthand derivative at a .

Remark 4.1

Assume that $f'(x_0)$ exists. The tangent line to the graph of $f(x)$ at $M(x_0; f(x_0))$ is

$$y = f(x_0) + f'(x_0)(x - x_0).$$

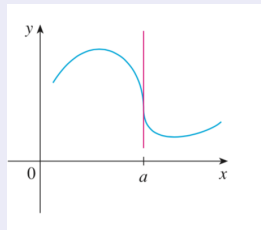
If $f'(x_0) \neq 0$ then the normal line to the graph of $f(x)$ at $M(x_0; f(x_0))$ is

$$y = f(x_0) - \frac{1}{f'(x_0)}(x - x_0).$$

Remark 4.2

If $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = \infty$,

the tangent line to the graph of $f(x)$ at $(x_0; f(x_0))$ is perpendicular to Ox .



Example 26

- ❶ $f(x) = 3^x$, compute $f'(2)$ and $f'(x)$.
- ❷ Compute $f'(0)$, where $f(x) = \begin{cases} \frac{1 - \cos 2x}{\ln(1+x)} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$
- ❸ Compute $h'(1)$, where $h(x) = \begin{cases} x^2 - 5x + 4 & \text{if } x \geq 1, \\ 2^x - 2 & \text{if } x < 1. \end{cases}$
- ❹ Compute $g'_+(0)$, where $g(x) = \begin{cases} x^{\frac{3}{2}} \sin \frac{1}{x} & \text{if } x > 0, \\ 0 & \text{if } x = 0. \end{cases}$

$$g'_+(0) = \lim_{x \rightarrow 0^+} \sqrt{x} \sin \frac{1}{x} = 0.$$

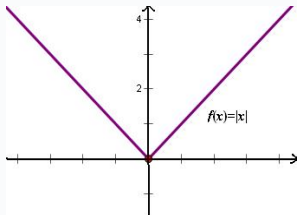
Connection between differentiability and continuity

- If f is differentiable at x , then f is continuous at x .

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x) - f(x_0)}{\Delta x}$$

$$\Rightarrow f(x_0 + \Delta x) - f(x_0) = f'(x_0)\Delta x + o(\Delta x) \xrightarrow{\Delta x \rightarrow 0} 0.$$

- The converse is wrong. For instance, $f(x) = |x|$, the function is continuous at $x = 0$ but is not differentiable at $x = 0$.



Theorem 4.1

Let $f(x), g(x)$ be defined and differentiable on (a, b) . Then $f(x) \pm g(x)$, $f(x).g(x)$, $\frac{f(x)}{g(x)}$ are also differentiable on (a, b) and

- $[f(x) \pm g(x)]' = f'(x) \pm g'(x)$.
- $[f(x).g(x)]' = f'(x)g(x) + f(x)g'(x)$.
- $[\frac{f(x)}{g(x)}]' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$, at $g(x) \neq 0$.
- *The chain rule:* $(f \circ g)'(x) = f'[g(x)].g'(x)$.

Example 27

$$(\sinh u(x))' = u'(x) \cosh u(x); (\cosh u(x))' = u'(x) \sinh u(x).$$

Theorem 4.2 (Derivative of the inverse function)

Assume that $f(x): (a, b) \rightarrow (c, d)$ has an inverse $f^{-1}(x): (c, d) \rightarrow (a, b)$. If $f(x)$ is differentiable at $x_0 \in (a, b)$, $f'(x_0) \neq 0$ and $f^{-1}(x)$ is continuous at $y_0 = f(x_0)$, then the inverse function $f^{-1}(x)$ is also differentiable at y_0 . Moreover,

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

Example 28

- 1 Find the derivative of $y = \arcsin x$.
- 2 Find the derivative of $y = \arccos x, y = \arctan x, y = \operatorname{arccot} x$.

$$C' = 0$$

$$(e^x)' = e^x$$

$$(\ln x)' = \frac{1}{x}$$

$$(\sin x)' = \cos x$$

$$(\tan x)' = \frac{1}{\cos^2 x}$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

$$(x^\alpha)' = \alpha x^{\alpha-1}, \alpha \in \mathbb{R}$$

$$(a^x)' = a^x \ln a, 0 < a \neq 1$$

$$(\log_a x)' = \frac{1}{x \ln a}, 0 < a \neq 1$$

$$(\cos x)' = -\sin x$$

$$(\cot x)' = -\frac{1}{\sin^2 x}$$

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$$

$$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$$

Example 29

Find the derivatives of the following functions

- ❶ $f(x) = 5 \sin(2x - 3) + 3 \sinh(5x).$
- ❷ $f(x) = (3x + 2) \arccos x - 7 \arctan(x^2 + 3).$
- ❸ $f(x) = \frac{\ln(x^2 + 4)}{x} - 8 \cot(xe^{2x}).$

Example 30

Let $f(x) = \ln(1 + x^2) + 2x + 2$ be a function whose inverse function is $g(x)$. Compute $g'(2)$.

Definition 27

- A function $y = f(x)$ is called an *explicit function*.
- Given an equation $F(x, y) = 0$. Any function $y = f(x)$ which satisfies $F(x, f(x)) = 0$ is called an *implicit function*.

Example 31

- $F(x, y) = x^2 + y^2 - 9 = 0$ defines two functions $y_{1,2} = \pm\sqrt{9 - x^2}$.
- In general, we can not solve for y from the equation $F(x, y) = 0$.

Example 32

Assume that the equation $x^3y - y^3x = 6$ defines an implicit functions $y = y(x)$. Compute $y'(x)$ and the equation of the tangent line at $A(2; 1)$.

We have $F(x, y(x)) = 0 \Leftrightarrow x^3y(x) - y^3(x)x = 6$.

Differentiating both sides (with respect to x), we gain

$$3x^2y(x) + x^3y'(x) - 3y^2(x)y'(x)x - y^3(x) = 0$$

$$\Rightarrow y'(x) = \frac{3x^2y(x) - y^3(x)}{3y^2x - x^3}.$$

At $A(2; 1)$: $x = 2$; $y(2) = 1$, we finally get $y'(2) = -\frac{11}{2}$.

The tangent line is $y = -\frac{11}{2}x + 12$.

Definition 28 (Higher order derivatives)

- *Second derivative:* $f''(x) = [f'(x)]'$.
- *The n -th derivative:* $f^{(n)}(x) = \left(f^{(n-1)}(x)\right)', n \geq 1$.

Convention: $f^{(0)}(x) = f(x)$.

Example 33

- $(e^{ax})^{(n)} = a^n e^{ax}$.
- $(\sin x)^{(n)} = \sin\left(x + n\frac{\pi}{2}\right)$.
- $(\cos x)^{(n)} = \cos\left(x + n\frac{\pi}{2}\right)$.
- $((ax + b)^\alpha)^{(n)} = a^n \cdot \alpha(\alpha - 1) \dots (\alpha - n + 1)(ax + b)^{\alpha - n}$.

Theorem 4.3 (Linearity)

$$\left(Af(x) + Bg(x)\right)^{(n)} = Af^{(n)}(x) + Bg^{(n)}(x),$$

A, B are constants.

Theorem 4.4 (Leibniz rule)

$$(f(x).g(x))^{(n)} = \sum_{k=0}^n C_n^k f^{(k)}(x)g^{(n-k)}(x).$$

Note: Leibniz rule should be applied when a factor is a polynomial.

Example 34

Compute the n -th derivatives of the functions

❶ $f(x) = \sin x \sin 3x.$

❷ $f(x) = \frac{1}{x^2 - 3x + 2}.$

❸ $f(x) = x^2 \sin(5x).$

❹ $f(x) = (2x + 1) \cos(3x).$

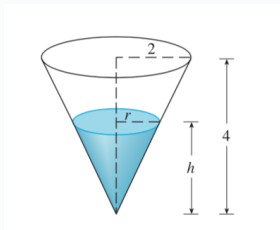
Example 35

Let the equation $F(x, y) = x + 2y - e^{xy} - 3 = 0$ define an implicit function $y(x)$. Find $y''(x)$ at $M(0; 2)$.

In a related rates problem, the idea is to compute the rate of change of one quantity in terms of the rate of change of other quantities (related quantities).

Example 36

A water tank has the shape of an inverted circular cone with base radius 2m and height 4m. If water is being pumped into the tank at a rate of $2\text{m}^3/\text{min}$, find the rate at which the water level is rising when the water is 3m deep?



Denote the water level by $h(t)$, the radius of the circular base at that level by $r(t)$, and the water volume in the tank by $V(t)$ at a certain time t . We have $V(t) = \frac{\pi}{3}r^2(t)h(t)$.

Moreover, $\frac{r}{h} = \frac{2}{4}$, so $V(t) = \frac{\pi}{12}h^3(t)$. That implies

$$V'(t) = \frac{\pi}{4}h^2(t).h'(t).$$

Hence, when the water is 3m deep and the rate of water being pumped is $2\text{m}^3/\text{min}$, the rising rate of water level is

$$h'(t) = \frac{4V'(t)}{\pi h^2(t)} = \frac{8}{9\pi}\text{m/min}.$$

Definition 29

Let $f(x): (a, b) \rightarrow \mathbb{R}$, $x_0 \in (a, b)$. If we can write

$$f(x_0 + \Delta x) - f(x_0) = A\Delta x + \alpha(\Delta x),$$

where $A \in \mathbb{R}$ and $\alpha(\Delta x) = o(\Delta x)$ as $\Delta x \rightarrow 0$, we say that $f(x)$ is differentiable at x_0 and the differential of $f(x)$ at x_0 is

$$df(x_0) = A\Delta x.$$

$f(x)$ has derivative at $x_0 \Leftrightarrow f(x)$ is differentiable at x_0 and $A = f'(x_0)$.

Choose $f(x) \equiv x$, we have $df(x) = dx = \Delta x$.

Hence, $df(x_0) = f'(x_0)dx$. df is a dependent variable, which depends on x and dx .

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0) \cdot \Delta x$$

Example 37

Approximate the following values $\sin 62^\circ$, $\sqrt[3]{\frac{2+0,06}{2-0,06}}$.

Given a differentiable function $y = f(x)$, we have

$$dy = f'(x)dx.$$

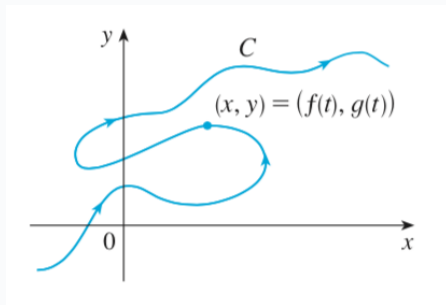
Assume that x is a dependent variable, namely $x = g(t) \Rightarrow y = f(g(t))$.

We write in terms of t

$$dy = (f \circ g)'(t)dt = f'(g(t))g'(t)dt = f'(x)dx.$$

The differential $dy = f'(x)dx$ is invariant, whether x is an independent or a dependent variable.

Parametric curves



It is impossible to describe this curve by an equation $y = y(x)$.

Assume that $f(t), g(t)$ are functions of the third variable, the **parameter** t . For each t , we determine a point $M(f(t), g(t))$. When t varies, M also varies and traces out a **parametric curve** C .

Assume that $f'(t) \neq 0 \forall t \in (a, b)$, then x has an inverse: $t = f^{-1}(x)$, we can rewrite $y = g(f^{-1}(x))$. It is obvious that $y = y(x)$ is differentiable and if $x'(t)$

$$y'(x) = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)}.$$

If $\frac{dx}{dt} = 0$, $\left(\frac{dy}{dt} \neq 0\right)$, the tangent is horizontal.

If $\frac{dy}{dt} = 0$, $\left(\frac{dx}{dt} \neq 0\right)$, the tangent is vertical.

Example 38

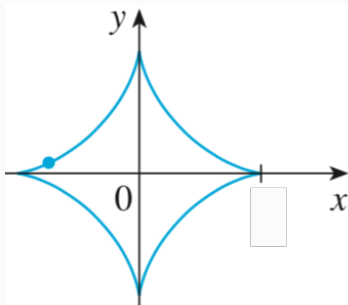
Compute the derivative $y'(x)$ of a function given by

$$a) \begin{cases} x = e^{t^2}, \\ y = te^{t^2}. \end{cases}$$

$$b) \begin{cases} x = t - e^t, \\ y = 2t + e^{-t^2}. \end{cases}$$

Example 39

Sketching the curve $\begin{cases} x = a \cos^3 t \\ y = a \sin^3 t. \end{cases}$

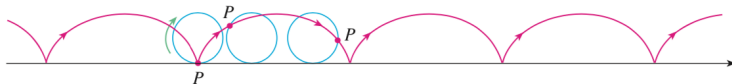


In Cartesian coordinates: $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}.$

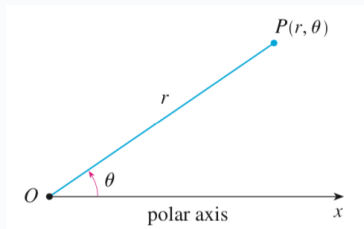
Example 40

The equation of a cycloid $\begin{cases} x = r(t - \sin t) \\ y = r(1 - \cos t) \end{cases}$.

EXAMPLE 7 Deriving parametric equations for a cycloid The curve traced out by a point P on the circumference of a circle as the circle rolls along a straight line is called a **cycloid** (see Figure 13). If the circle has radius r and rolls along the x -axis and if one position of P is the origin, find parametric equations for the cycloid.



Polar coordinates



Choose in the plane a fixed point O called the **pole**.

Draw a ray starting at O called the **polar axis**. A point P has **polar coordinates** (r, θ) determined as follows:

- $r = |\overrightarrow{OP}|$; $0 \leq r \leq \infty$ – **polar radius**,
- $\theta = (\overrightarrow{Ox}, \overrightarrow{OP})$; $0 \leq \theta < 2\pi$ (rotating $\overrightarrow{Ox}$ counterclockwise until reaching $\overrightarrow{OP}$) – **polar angle**.

The pole $O(0, \theta_0)$ for arbitrary θ_0 .

Converting to the Cartesian coordinates xOy :

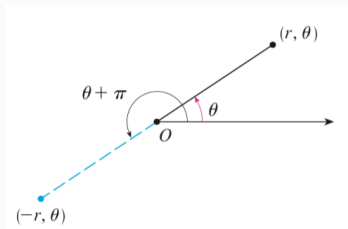
$$(r, \theta) \mapsto (x, y), x = r \cos \theta, y = r \sin \theta.$$

$$(x, y) \mapsto (r, \theta), r = \sqrt{x^2 + y^2}, \tan \theta = \frac{y}{x},$$

such that $\sin \theta$ and y are of the same sign.

Generalized polar coordinates

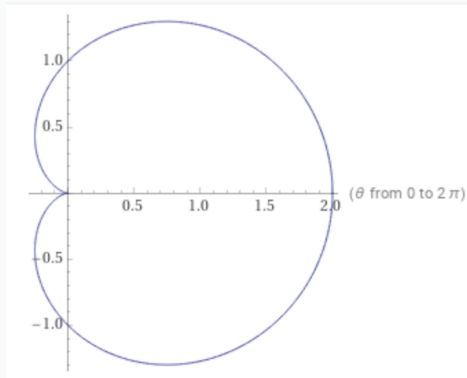
When mentioning a curve given in polar coordinates, one often means the generalized polar coordinates (r, θ) , where $r \in \mathbb{R}, \theta \in \mathbb{R}$, which corresponds to the following point in polar coordinates:

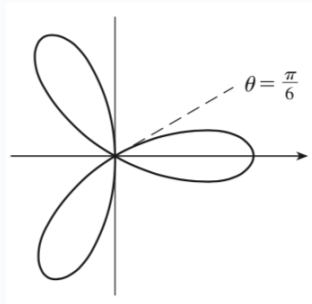
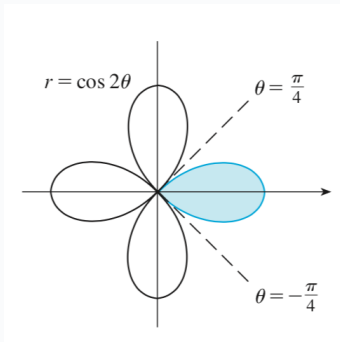


- If $r \geq 0$ then $(r, \theta) = (r, \theta_0)$ where $\theta_0 \in [0, 2\pi)$ and $\theta - \theta_0 = 2k\pi, k \in \mathbb{Z}$.
- If $r < 0$ then $(r, \theta) = (-r, \theta + \pi)$.

Example 41

Sketching the curve $r = 1 + \cos \theta$.





Hình: Four leaved rose $r = a \cos 2\theta$. Three leaved rose $r = a \cos 3\theta$, ($a > 0$).

Theorem 5.1

- $d(u \pm v) = du \pm dv.$
- $d(u.v) = vdu + u dv.$
- $d\left(\frac{u}{v}\right) = \frac{vdu - u dv}{v^2}, v \neq 0.$

Higher order differentials

Definition 30

Let $f(x)$ have derivatives up to the n -th order.

- The second differential:
$$d^2 f = d[df] = d[f'(x)dx] = d[f'(x)]dx = f''(x)(dx)^2.$$
- The n -th differential: $d^n f(x) = f^{(n)}(x)(dx)^n.$

Note: Higher order differentials depend on whether x is a dependent or an independent variable.

Example 42

- 1 $y = (2x + 3) \cos(2x)$. Calculate $d^{10}y(0)$.
- 2 $y = \frac{x^3}{1+x}$. Calculate $d^n y$, $n \geq 4$.

If $f(x)$ is differentiable, we have a linear approximation:

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0)\Delta x.$$

Aim: Let $f(x)$ have higher order derivatives. Can we approximate $f(x)$ when x is near x_0 by a polynomial

$$P_n(x) = c_0 + c_1(x - x_0) + \dots + c_n(x - x_0)^n,$$

that $f(x_0) = P_n(x_0)$, $f'(x_0) = P'_n(x_0)$, $\dots$, $f^{(n)}(x_0) = P_n^{(n)}(x_0)$.

We obtain $c_k = \frac{f^{(k)}(x_0)}{k!}$, $0 \leq k \leq n$.

Definition 31 (Taylor polynomials)

$$P_n(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \dots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

is called *the n th degree Taylor polynomial of $f(x)$ centered at x_0* .

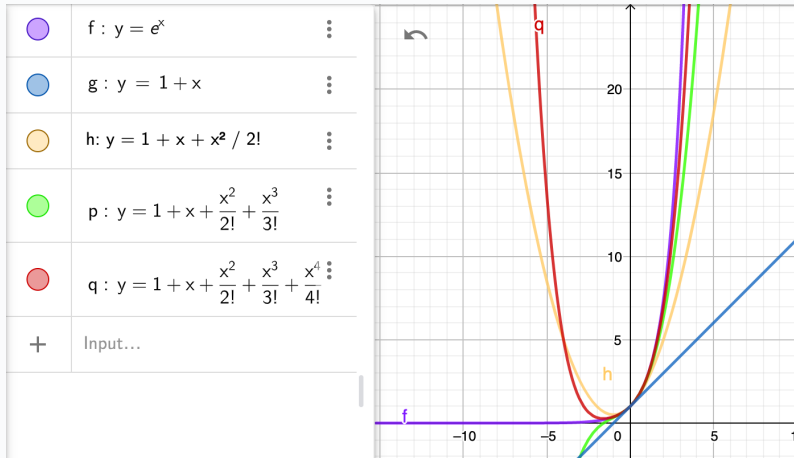
Later: Estimate the error $|f(x) - P_n(x)|$?

Example 43

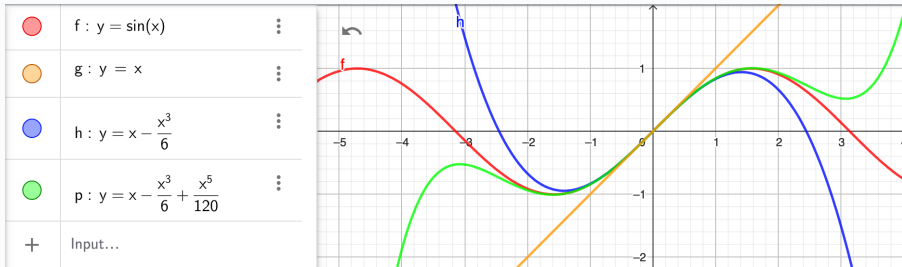
Write the Taylor polynomial of the following functions of given degree

- 1 $y = e^x, x_0 = 0.$
- 2 $y = \frac{1}{\sqrt{x}}, x_0 = 1, n = 3.$

- $y = e^x, x_0 = 0.$
- $y^{(n)}(x) = e^x, y^{(n)}(0) = 1.$
- $P_n(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^n}{n!}.$



- $y = \sin x, x_0 = 0.$
- $y^{(n)}(x) = \sin(x + \frac{n\pi}{2}).$
- $P_{2k+1}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots + (-1)^k \frac{x^{2k+1}}{(2k+1)!}.$



Theorem 5.2 (Taylor expansion)

Let $f(x)$ be $(n+1)$ times differentiable on (a, b) , with $f^{(n)}$ continuous on $[a, b]$. Then for all $x_0 \in (a, b)$, we have

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(\bar{x}_0)}{(n+1)!} (x - x_0)^{n+1},$$

for some real number $\bar{x}_0$ between x and x_0 .

The Taylor expansion at $x_0 = 0$ is called the **Maclaurin expansion** of $f(x)$.

Corollary 5.1

If $|f^{(n+1)}(x)| \leq M, \forall x \in (a, b)$, then

$$|R_n(x)| = |f(x) - P_n(x)| \leq \frac{M}{(n+1)!} |x - x_0|^{n+1}.$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \frac{e^{\bar{x}_0}}{(n+1)!} x^{n+1}$$

$$\sin x = x - \frac{x^3}{3!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \frac{\sin(\bar{x}_0 + (2n+2)\frac{\pi}{2})}{(2n+2)!} x^{2n+2}$$

$$\cos x = 1 - \frac{x^2}{2!} + \dots + \frac{(-1)^n x^{2n}}{(2n)!} + \frac{\cos(\bar{x}_0 + (2n+1)\frac{\pi}{2})}{(2n+1)!} x^{2n+1}$$

where $\bar{x}_0$ lies between 0 and x .

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots + \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}x^n + \\ + \frac{\alpha(\alpha-1)\dots(\alpha-n)}{(n+1)!}(1+\bar{x}_0)^{\alpha-n-1}x^{n+1},$$

$$\ln(1+x) = x - \frac{x^2}{2} + \dots + (-1)^{n-1}\frac{x^n}{n} + \frac{(-1)^n}{(n+1)(1+\bar{x}_0)^{n+1}}x^{n+1},$$

where $\bar{x}_0$ lies between 0 and x .

$$f(x) = P_n(x - x_0) + o((x - x_0)^n), \text{ as } x \rightarrow x_0,$$

$o((x - x_0)^n)$ is an infinitesimal of higher order than $(x - x_0)^n$.

Example 44 (Finite expansion of essential functions)

As $x \rightarrow 0$:

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + o(x^n)$$

$$\sin x = x - \frac{x^3}{3!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + o(x^{2n+2})$$

$$\cos x = 1 - \frac{x^2}{2!} + \dots + \frac{(-1)^n x^{2n}}{(2n)!} + o(x^{2n+1}).$$

Example 45 (Finite expansion of essential functions)

As $x \rightarrow 0$:

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \dots +$$

$$+ \frac{\alpha(\alpha-1)\dots(\alpha-n+1)}{n!}x^n + o(x^n),$$

$$\ln(1+x) = x - \frac{x^2}{2} + \dots + (-1)^{n-1} \frac{x^n}{n} + o(x^n).$$

Example 46

Write the expansion to the given order of the following functions

① $f(x) = \sin 2x + x \cos 3x^2$, $x_0 = 0$, $n = 5$.

② $g(x) = \frac{1}{\sqrt{x}}$, $x_0 = 1$, $n = 3$. Apply to approximate the value $f(1,01)$, and estimate the error.

Example 47

Compute the following limits

a) $\lim_{x \rightarrow 0} \frac{e^x \cdot \cos 2x - 1 - x}{x^2}$

b) $\lim_{x \rightarrow 0} \frac{\sqrt[3]{1+2x} - \sqrt[6]{1+4x}}{x^2}$

Theorem 5.3

Let $f(x)$ be defined on (a, b) and attain a local maximum or minimum at $c \in (a, b)$. If there exists $f'(c)$, then $f'(c) = 0$.

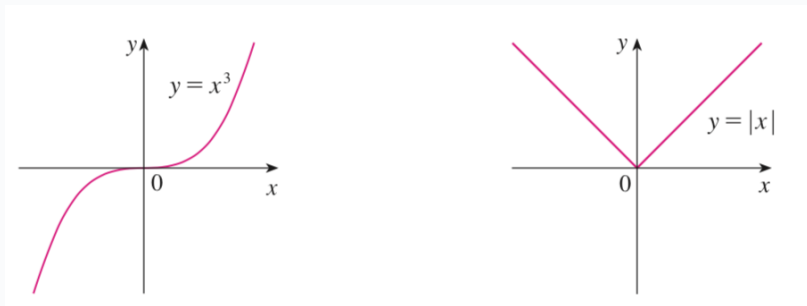
Without loss of generality, assume that c is a local minimum point of $f(x)$.

Then

$$f'_+(c) = \lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \geq 0$$

$$f'_-(c) = \lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \leq 0$$

There exists $f'(c)$, hence, $f'_+(c) = f'_-(c) = 0$.



- 1 At a local extreme point, it can happen that $f'(c)$ does not exist.
- 2 The converse of Fermat's theorem is not true. There can be a stationary point ($f'(x) = 0$) that is not a local extreme point.

Theorem 5.4

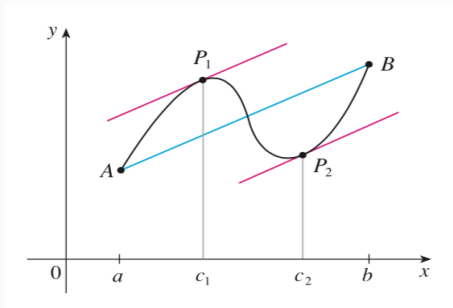
Let $f(x)$ be defined and continuous on $[a, b]$, differentiable on (a, b) . If $f(a) = f(b)$, then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Lagrange's theorem

Theorem 5.5

Let $f(x)$ be defined and continuous on $[a, b]$, differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}, \text{ or } f(b) - f(a) = f'(c)(b - a).$$



Hình: Geometric interpretation

Corollary 5.2

Let $f(x)$ be differentiable and $f'(x) = 0$ for $x \in (a, b)$. Then, $f(x) = C$ for all $x \in (a, b)$.

Proof by contradiction.

Theorem 5.6

Let $f(x)$, $g(x)$ be continuous on $[a, b]$, differentiable in (a, b) , and $g'(x) \neq 0$ on (a, b) . There exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Example 48

Is the Cauchy's theorem applicable to the functions $f(x) = x^2$ and $g(x) = x^3$ on $[-1, 3]$?

Remark: a mean value theorem can be inapplicable, but the formula can be fulfilled by direct calculation.

Example 49

- 1 Show that the equation $a \cos x + b \cos 2x + c \cos 3x = 0$ has at least one root on $(0, \pi)$.
- 2 Let a, b, c satisfy $a + b + c = 0$. Prove that the equation $ax^3 + 2bx + 2c = 0$ has a solution in $(0, 2)$.
- 3 Prove that $\frac{y-x}{1+y^2} < \arctan y - \arctan x < \frac{y-x}{1+x^2}$ for all $0 < x < y$.
- 4 Prove that $|\cos x - \cos y| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

Theorem 5.7 (L'Hospital's rule)

Suppose $f(x)$ and $g(x)$ are differentiable, $g'(x) \neq 0$ near x_0 , possibly except at x_0 . Assume that

$$\lim_{x \rightarrow x_0} f(x) = 0 \text{ and } \lim_{x \rightarrow x_0} g(x) = 0$$

or

$$\lim_{x \rightarrow x_0} f(x) = \infty \text{ and } \lim_{x \rightarrow x_0} g(x) = \infty,$$

and there exists $\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} \in \overline{\mathbb{R}}$. Then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}.$$

Consider $F(x) = \begin{cases} f(x) & \text{if } x \neq x_0, \\ 0 & \text{if } x = x_0; \end{cases} G(x) = \begin{cases} g(x) & \text{if } x \neq x_0, \\ 0 & \text{if } x = x_0. \end{cases}$

Applying Cauchy's theorem, there exists $c \in (x, x_0)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{F'(c)}{G'(c)} = \frac{F(x) - F(x_0)}{G(x) - G(x_0)} = \frac{f(x)}{g(x)}.$$

Passing to the limit as $x \rightarrow x_0$, then $c \rightarrow x_0$, we obtain the desired equality.

- The rule holds for one sided limits, limit at infinity.
- This rule is a sufficient condition to evaluate the indeterminate forms, consider $\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \sin x}$.

Indeterminate form $\frac{\infty}{\infty}$. And $\lim_{x \rightarrow \infty} \frac{x - \sin x}{x + \sin x} = \lim_{x \rightarrow \infty} \frac{x}{x} = 1$.

However, $\lim_{x \rightarrow \infty} \frac{1 - \cos x}{1 + \cos x} = \lim_{x \rightarrow \infty} \tan^2 \frac{x}{2}$ does not exist.

Example 50

Find the limits

$$① \lim_{x \rightarrow 0} \frac{e^x \sin x - x(1+x)}{x^3}$$

$$② \lim_{x \rightarrow 1^-} \frac{\tan \frac{\pi}{2}x}{\ln(1-x)}$$

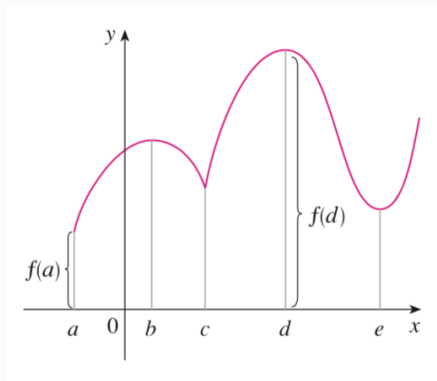
$$③ \lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$$

$$④ \lim_{x \rightarrow 0} (3^x + 4^x - 5^x)^{\frac{1}{3^x + 4^x - 2 \cdot 5^x}}$$

Definition 32

Let $f(x)$ be defined on D . $f(x)$ has

- *an absolute maximum* at $c \in D$ if $f(c) \geq f(x) \forall x \in D$. $M = f(c)$ is called the *maximum value* of f on D .
- *an absolute minimum* at $d \in D$ if $f(d) \leq f(x) \forall x \in D$. $m = f(d)$ is called the *minimum value* of f on D .
- *a local maximum* at $c \in D$ if $f(c) \geq f(x), \forall x \in (c - \varepsilon, c + \varepsilon)$. $f(c)$ is called a *local maximum value* of f on D .
- *a local minimum* at $d \in D$ if $f(d) \leq f(x), \forall x \in (d - \varepsilon, d + \varepsilon)$. $f(d)$ is called a *local minimum value* of f on D .



Corollary 5.3 (A necessary condition for attaining extreme values)

If $f(x)$ has a local maximum or minimum at c then c is a **critical point of $f(x)$** , i.e., either $f'(c) = 0$ or $f'(c)$ does not exist.

Theorem 5.8 (First Derivative Test)

Suppose that c is a critical number of a continuous function f .

- If $f'(x)$ changes from positive to negative at c , then f has a local maximum at c .
- If $f'(x)$ changes from negative to positive at c , then f has a local minimum at c .
- If $f'(x)$ does not change sign at c , then f has no local extremum at c .

Theorem 5.9 (Second Derivative Test)

Suppose that $f'(c) = 0$ and $f''(x)$ is continuous at c .

- If $f''(c) > 0$ then f has a local minimum at c .
- If $f''(c) < 0$ then f has a local maximum at c .

Theorem 5.10

If $f(x)$ is continuous on $[a, b]$ then it attains its minimum and maximum on $[a, b]$.

- 1 Determine critical points $c_i \in (a, b), i = 1, 2, \dots, n$.
- 2 $\max_{x \in [a, b]} f = \max\{f(c_i), f(a), f(b)\}, \quad \min_{x \in [a, b]} f = \min\{f(c_i), f(a), f(b)\}.$

Example 51

- 1 Find the local extreme values of $f(x) = x^{2/3}(x + 2)$.
- 2 A rectangular plot of farmland will be bounded on one side by a river and in the other three sides by a single-strand electric fence. With 800m of wire at your disposal, what is the largest area you can enclose, and what are its dimensions?
- 3 What are the dimensions of the lightest open-top right circular cylindrical can that will hold a volume of 1000cm^3 ?

Theorem 5.11 (Increasing/Decreasing Test)

- If $f'(x) > 0$ on an interval then $f(x)$ is increasing on that interval.
- If $f'(x) < 0$ on an interval then $f(x)$ is decreasing on that interval.

Proof. Applying Lagrange's theorem.

Definition 33

A point $I(c, f(c))$ is called *an inflection point of $y = f(x)$* iff $f''(x)$ changes its sign when crossing $x = c$.

Theorem 5.12 (Second Derivative Test)

Suppose that $f(x)$ is twice differentiable.

- If $f''(x) > 0$ on (a, b) then f is concave upward on that interval.
- If $f''(x) < 0$ on (a, b) then f is concave downward on that interval.

- ① $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ iff $\lim_{x \rightarrow a} y = \infty$.
- ② $y = b$ is called a **horizontal asymptote** of the curve $y = f(x)$ iff $\lim_{x \rightarrow \pm\infty} y = b$.
- ③ $y = ax + b$ is called a **slant (oblique) asymptote** of the curve $y = f(x)$ iff $\lim_{x \rightarrow \pm\infty} [y - ax - b] = 0$. We have
$$a = \lim_{x \rightarrow \pm\infty} \frac{y}{x}, b = \lim_{x \rightarrow \pm\infty} [y - ax].$$

Example 52

Determine the asymptotes of the following curves

$$\textcircled{1} \quad y = \frac{x^2 + 3x - 1}{x + 1}.$$

$$\textcircled{2} \quad y = \frac{xe^x}{x + 1}.$$

$$\textcircled{3} \quad y = x \operatorname{arccot} \frac{2}{x}.$$

$$\textcircled{4} \quad x = \frac{t^2}{t - 1}, \quad y = \frac{t}{t^2 - 1}.$$